

YU ZHONG, China

WMO: 529830

Lat: 35.8664N

Lon: 104.1431E

Elev: 1875

StdP: 80.74

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-14.4	-21.8	0.7	-10.5	-20.0	0.8	-9.0	5.5	-3.2	4.9	-3.7	1.3	250	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	29.4	16.4	27.9	16.2	26.5	15.6	19.1	25.3	18.4	24.2	17.7	23.2	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	15.7	21.8	16.6	14.9	21.4	15.8	14.2	20.6	63.4	25.3	60.9	24.3	58.2	23.2	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.5	4.8	-19.1	32.0	3.1	1.4	-21.3	33.1	-23.1	33.9	-24.8	34.7	-27.0	35.8
			-19.6	20.2	2.9	0.9	-21.7	20.9	-23.4	21.4	-25.0	21.8	-27.0	22.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-6.4	-2.2	4.4	10.3	14.1	17.8	19.8	19.1	14.2	8.7	1.5	-4.9
	DBStd	9.59	3.57	4.26	4.16	3.82	3.31	2.39	2.52	2.73	2.59	3.17	3.65	3.10
	HDD10.0	1854	507	342	178	41	6	0	0	0	2	63	255	461
	HDD18.3	3869	766	575	432	239	133	37	14	23	126	300	505	719
	CDD10.0	1157	0	0	4	51	134	233	303	282	128	22	0	0
	CDD18.3	131	0	0	0	0	3	20	59	47	2	0	0	0
	CDH23.3	1323	0	0	0	18	78	246	563	395	22	2	0	0
	CDH26.7	278	0	0	0	2	7	38	147	83	1	0	0	0
Wind	WSAvg	2.1	1.7	1.9	2.3	2.5	2.5	2.4	2.4	2.2	2.0	2.0	1.7	1.6
Precipitation	PrecAvg	352	3	4	10	21	47	50	71	69	48	25	3	1
	PrecMax	502	10	12	28	83	99	137	145	134	86	56	11	4
	PrecMin	243	0	0	0	3	12	18	16	22	18	1	0	0
	PrecStd	58	2	3	6	16	26	28	34	30	21	12	3	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	13.6	20.5	25.8	27.5	29.5	32.2	30.9	25.6	21.6	14.7	7.5
		MCWB	-0.2	3.9	7.5	10.5	12.7	14.8	17.3	17.6	14.9	11.2	5.5	0.6
	2%	DB	4.9	9.7	17.0	22.6	25.4	27.8	29.8	28.9	23.5	19.1	12.5	4.9
		MCWB	-1.9	1.7	6.0	9.5	12.0	14.6	16.7	17.3	14.0	10.6	4.8	-0.8
	5%	DB	2.5	7.3	14.5	20.2	23.7	26.2	28.3	27.3	21.7	16.9	10.3	3.2
		MCWB	-3.1	0.6	4.8	8.7	11.7	14.3	16.7	16.9	13.7	9.8	4.0	-1.7
	10%	DB	0.3	4.9	12.1	18.1	21.8	24.5	26.6	25.6	20.0	14.8	8.3	1.3
		MCWB	-4.3	-0.5	3.8	7.7	11.2	14.0	16.5	16.5	13.5	8.7	3.0	-2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	5.3	9.2	12.4	15.4	18.3	20.4	19.9	17.7	13.1	6.7	1.1
		MCDB	7.6	12.3	17.6	20.9	22.9	24.1	28.1	27.1	21.4	18.4	12.5	6.0
	2%	WB	-1.5	2.5	6.8	10.7	14.1	17.2	19.4	19.1	16.1	11.8	5.5	-0.4
		MCDB	4.6	8.6	14.5	19.1	20.8	22.7	25.9	25.1	20.0	16.5	10.6	4.2
	5%	WB	-2.9	0.9	5.4	9.7	13.1	16.2	18.6	18.5	15.1	10.8	4.6	-1.5
		MCDB	2.2	6.7	13.2	18.1	20.1	21.7	24.6	24.2	18.9	15.2	9.6	2.8
	10%	WB	-4.1	-0.5	4.2	8.6	12.2	15.4	17.8	17.9	14.1	9.7	3.3	-2.6
		MCDB	0.0	4.4	11.4	16.3	19.6	21.3	23.6	23.1	18.3	13.9	7.9	1.1
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.8	11.7	12.6	12.3	11.9	11.7	11.1	10.0	10.6	11.3	11.5
		MCDBR	15.7	16.0	16.3	16.7	16.2	16.1	15.7	14.2	14.0	14.3	14.8	13.8
		MCWBR	10.2	9.3	7.6	7.0	6.3	5.9	5.3	4.9	5.9	7.1	8.2	8.9
	5% WB	MCDBR	15.1	15.1	14.8	14.6	13.0	11.4	11.8	10.9	10.2	12.0	13.3	12.9
		MCWBR	9.9	9.0	7.2	6.6	5.7	4.9	4.7	4.2	5.2	6.5	7.8	8.4
Clear-Sky Solar Irradiance	taub	0.346	0.403	0.465	0.499	0.494	0.479	0.462	0.474	0.451	0.418	0.357	0.327	
	taud	2.094	1.949	1.825	1.784	1.875	1.979	2.105	2.040	2.053	2.050	2.131	2.177	
	Ebn at Noon	822	811	794	791	803	813	824	805	801	786	803	821	
	Edh at Noon	121	159	197	217	201	181	158	166	155	142	116	105	
All-Sky Solar Radiation	RadAvg	2.88	3.64	4.65	5.52	5.91	5.83	5.69	5.23	3.98	3.61	3.09	2.77	
	RadStd	0.29	0.28	0.32	0.33	0.36	0.34	0.54	0.38	0.30	0.34	0.25	0.17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page